

Hash Table Exercises

Instructions

Complete the following tasks using Python dictionaries.
Each task introduces key concepts related to hash tables.

Learning Objectives

- Understand basic hash table operations
 - Practice dictionary manipulation
 - Learn about collisions and collision handling
 - Observe hash function behavior
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Task 1: Student Management System

Create a dictionary to store student information (ID → Name).

1. Add 3 students
 2. Print the name of a specific student
 3. Update a student's name
 4. Remove a student
 5. Print all students
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Task 2: Grade Calculator

Create a dictionary to store subject grades.

1. Add grades for Math, Physics, and Chemistry
 2. Calculate the average grade
 3. Find the highest and lowest grade
 4. Add a new subject with its grade
 5. Update an existing subject grade
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Task 3: Simple Hash Function

1. Create a hash function that maps strings to numbers from 0 to 9
 2. Test the function using 5 different names
 3. Count how many collisions occur
 4. Print the names that share the same hash value
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Task 4: Collision Handling

Create a hash table that uses lists to handle collisions.

1. Implement an insert operation
2. Implement a search operation
3. Insert 5 items that cause collisions
4. Print all items stored in each bucket